Largest induced trees in graphs of given girth and maximum degree, with a proof of Graffiti.pc's Conjecture 141

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Abstract

For a finite simple graph G, let $\mathrm{t}(G)$ denote the largest number of vertices inducing a tree, $\mathrm{g}(G)$ the girth, $\Delta(G)$ the maximum degree, and for a vertex v let $\ell(v)$ denote the independence number of the subgraph induced by the neighbourhood of v. We prove that every finite connected triangle-free graph containing a cycle satisfies $\mathrm{t}(G) \geq \Delta(G)+\mathrm{g}(G)-3$, with equality for all cycles C_g with $g \geq 4$ and all complete bipartite graphs $K_{a,b}$ with $a, b \geq 2$. As a corollary we prove Conjecture 141 of the conjecture-making program Graffiti.pc (DeLaViña, *Written on the Wall II*, 2005): every finite connected graph satisfies $\mathrm{t}(G) \geq \lfloor \mathrm{g}(G)/2 \rfloor - 1 + \max_v \ell(v)$. The proofs are elementary maximality arguments. Both results have been formalized and machine-checked in Lean 4 against the statement of the conjecture in the Google DeepMind *Formal Conjectures* repository.

1 Introduction

Graffiti.pc, developed by DeLaViña [1, 2] as a successor of Fajtlowicz's program Graffiti [5], generates conjectural inequalities between graph invariants; its conjectures are collected in the lists *Written on the Wall II* [3], with a curated register maintained by West [4]. A run of these conjectures bounds from below the invariant $\mathrm{t}(G)$, the largest order of an induced subtree, whose systematic study goes back to Erdős, Saks, and Sós [6].

Throughout, G is a finite simple graph, $\mathrm{g}(G)$ its girth, $\Delta(G)$ its maximum degree, and for a vertex v we write $\ell(v)$ for the independence number of the subgraph induced by the neighbourhood $N(v)$. Conjecture 141 of Written on the Wall II, dated 2005, asserts [3]:

If G is a simple connected graph, then $\mathrm{t}(G) \;\geq\; \lfloor \mathrm{g}(G)/2 \rfloor - 1 + \max_v \ell(v)$.

At the time of writing the conjecture is listed as open on DeLaViña's register (status "O") [3] and is stated as a research-open item in the Google DeepMind *Formal Conjectures* repository [10, 11]. We prove it. The main step is the following sharper bound, which does not involve ℓ at all and may be of independent interest.

Theorem 1.1. *Let G be a finite simple connected triangle-free graph that contains a cycle. Then*

$$\mathrm{t}(G) \;\geq\; \Delta(G) + \mathrm{g}(G) - 3.$$

Equality holds for every cycle C_g ($g \geq 4$) and every complete bipartite graph $K_{a,b}$ with $a, b \geq 2$.

Corollary 1.2 (Conjecture 141)**.** *Every finite simple connected graph G on at least two vertices satisfies* $\mathrm{t}(G) \geq \lfloor \mathrm{g}(G)/2 \rfloor - 1 + \max_v \ell(v)$*, where* $\mathrm{g}(G) := 0$ *if G is acyclic.*

The deduction of the corollary is short: for girth ≤ 5 (and for acyclic graphs) an induced star at a vertex maximizing ℓ already beats the bound, while for girth ≥ 6 the graph is triangle-free, so $\ell(v) = \deg(v)$ for every v, and Theorem 1.1 applies with $\mathrm{g} - 3 \geq \lfloor \mathrm{g}/2 \rfloor - 1$.

The proof of Theorem 1.1 is a maximality argument in the spirit of our proof of Graffiti.pc's Conjecture 143 [9]: a maximum-order induced tree containing the closed neighbourhood of a maximum-degree vertex either spans (impossible in a cyclic graph) or admits an outside vertex with two neighbours on the tree; the tree path joining those two neighbours closes a cycle, hence has length at least $\mathrm{g}(G) - 2$, and it can share at most three vertices with the closed neighbourhood.

Related work. The foundational paper [6] bounds $\mathrm{t}(G)$ in terms of order, size, radius, and independence and clique numbers; we are not aware of a published lower bound for t combining girth with the maximum degree or with neighbourhood independence, and the closest Graffiti.pc antecedents (the induced-forest bounds of DeLaViña–Waller [7]; see also Hertz–Marcotte–Schindl [8]) concern induced forests rather than trees. DeLaViña's resolved-conjecture lists do not contain Conjecture 141 [3]. As with any short elementary argument, we do not claim it could not have been observed before.

Verification. Theorem 1.1 and Corollary 1.2 have been formalized and machine-checked in Lean 4 [12] on top of Mathlib [13], against the pre-existing formal statement `conjecture141` of the Formal Conjectures repository [11]; see Section 6. Independently, the conjecture and all intermediate claims were verified exhaustively by exact computation on all 995 connected graphs on 2–7 vertices and on structured families (cages, prisms, generalized theta graphs, cycles with pendant trees, complete bipartite graphs, and seeded random graphs); no violation was found, and every equality case located has girth exactly 4. The computations play no role in the proofs.

2 Notation

All graphs are finite and simple. For $S \subseteq V(G)$, $G[S]$ is the induced subgraph; S *induces a tree* if $G[S]$ is connected and acyclic, and $\mathrm{t}(G) = \max\{|S| : G[S] \text{ is a tree}\}$. The girth of a graph containing a cycle is the minimum length of a cycle; following the Mathlib convention used by the formal statement, an acyclic graph has girth 0. $N(v)$ and $N[v] = N(v) \cup \{v\}$ are the open and closed neighbourhoods; $\ell(v)$ is the independence number of $G[N(v)]$. Note $\ell(v) \leq \deg(v)$ always, with equality when $N(v)$ is independent, i.e. for all v when G is triangle-free.

3 The star lemma

Lemma 3.1. *For every vertex v of any graph G:* $\mathrm{t}(G) \geq \ell(v) + 1$.

Proof. Let $S \subseteq N(v)$ be independent with $|S| = \ell(v)$. Then $G[\{v\} \cup S]$ is a star: all edges vs ($s \in S$) are present and there are no edges inside S. A star is a tree. $\square$

4 Proof of Theorem 1.1

Proof of Theorem 1.1. Write $g = \mathrm{g}(G) \geq 4$ and $\Delta = \Delta(G)$. Fix a vertex v of degree Δ. Since G is triangle-free, $N(v)$ is independent, so by the proof of Lemma 3.1 the closed neighbourhood $N[v]$ induces a star, hence a tree. Among all sets $S \supseteq N[v]$ inducing a tree choose one, say S^*, of maximum cardinality, and let $T = G[S^*]$.

S^* is a proper subset of $V(G)$: an induced spanning tree would make G itself a tree, contradicting the existence of a cycle. Since G is connected, some vertex $z \notin S^*$ has a neighbour in S^*; and z must have two distinct neighbours $a, b \in S^*$, for otherwise $G[S^* \cup \{z\}]$ would be the tree T with a pendant vertex attached, an induced tree containing $N[v]$ and larger than T — contradicting maximality.

Let P be the unique a–b path in T. The edges of P together with za and zb form a cycle of length $|E(P)| + 2$, so

$$|E(P)| \;\geq\; g - 2. \tag{4.1}$$

Next, $|V(P) \cap N[v]| \leq 3$. Every edge of G between two vertices of S^* is an edge of the tree T (as T is induced), and paths between fixed endpoints in a tree are unique. If $v \notin V(P)$ and P contained two distinct vertices $u_1, u_2 \in N(v)$, then the segment of P between u_1 and u_2 would be the unique u_1–u_2 path in T; but $u_1 v u_2$ is also a u_1–u_2 path in T, so the segment would pass through v, contradicting $v \notin V(P)$; hence $|V(P) \cap N[v]| \leq 1$ in this case. If $v \in V(P)$, then for any $u \in V(P) \cap N(v)$ the segment of P between v and u is the unique v–u path of T, which is the single edge vu; so u is adjacent to v *along* P, and a path has at most two edges at any vertex: $|V(P) \cap N(v)| \leq 2$, so $|V(P) \cap N[v]| \leq 3$.

Finally, count. $V(P)$ and $N[v]$ are subsets of S^*, so by inclusion–exclusion and (4.1),

$$|S^*| \;\geq\; |V(P) \cup N[v]| \;=\; |V(P)| + |N[v]| - |V(P) \cap N[v]| \;\geq\; (g-1) + (\Delta+1) - 3 \;=\; \Delta + g - 3.$$

Since T is an induced tree, $\mathrm{t}(G) \geq |S^*| \geq \Delta + g - 3$.

For sharpness: in C_g ($\Delta = 2$) the largest induced trees are the paths obtained by deleting one vertex, so $\mathrm{t} = g - 1 = \Delta + g - 3$; in $K_{a,b}$ with $a \leq b$ ($g = 4$, $\Delta = b$) every induced tree is a star (two vertices on each side already induce C_4), so $\mathrm{t} = b + 1 = \Delta + g - 3$. $\square$

Remark 4.1. Under the hypotheses of Theorem 1.1, an alternative proof of the slightly weaker bound $\mathrm{t} \geq \Delta + \lfloor g/2 \rfloor - 1$ (still sufficient for Corollary 1.2) is worth recording. For $r = \lfloor g/2 \rfloor - 1$, the ball $B(v, r)$ induces a tree in any graph of girth g: every G-edge inside the ball is an edge of a breadth-first search tree rooted at v, since a non-tree edge would close a cycle of length at most $2r + 1 < g$ through the last common ancestor of its endpoints. Now take v of maximum degree; triangle-freeness gives $g \geq 4$, so $r \geq 1$. If G contains a cycle the ball is proper, so all r distance layers are nonempty and $|B(v, r)| \geq 1 + \Delta + (r - 1)$.

5 Proof of Corollary 1.2

Proof of Corollary 1.2. Let $L = \max_v \ell(v)$; note $L \geq 1$ since G is connected on ≥ 2 vertices. If G is acyclic then $\mathrm{g} = 0$ and the claimed bound is $L - 1 \leq \mathrm{t}$, which follows from Lemma 3.1. If $3 \leq \mathrm{g} \leq 5$ then $\lfloor \mathrm{g}/2 \rfloor - 1 \leq 1$ and Lemma 3.1 again gives $\mathrm{t} \geq L + 1 \geq \lfloor \mathrm{g}/2 \rfloor - 1 + L$. If $\mathrm{g} \geq 6$ then G is triangle-free, so $\ell(v) = \deg(v)$ for every v and $L = \Delta$; by Theorem 1.1, $\mathrm{t} \geq \Delta + \mathrm{g} - 3 \geq \Delta + \lfloor \mathrm{g}/2 \rfloor - 1 = L + \lfloor \mathrm{g}/2 \rfloor - 1$, using $\mathrm{g} - 3 \geq \lfloor \mathrm{g}/2 \rfloor - 1$ for $\mathrm{g} \geq 4$. $\square$

Remark 5.1. The corollary is sharp precisely in girth 4. Equality holds for C_4, for every $K_{a,b}$ ($a, b \geq 2$), and for the infinite family obtained from C_4 by attaching $k \geq 0$ pendant vertices to one vertex ($\ell_{\max} = k + 2$, $\mathrm{t} = k + 3$); an exhaustive check of all connected graphs on at most seven vertices finds no equality case of any other girth. That equality *requires* girth 4 follows from the results above: for acyclic G and for $g = 3$ the star bound $\mathrm{t} \geq L + 1$ beats the right-hand side by at least 2 resp. 1; for $g = 5$ triangle-freeness gives $L = \Delta$ and Theorem 1.1 yields $\mathrm{t} \geq \Delta + 2 = L + \lfloor 5/2 \rfloor$, one more than required; and for $g \geq 6$ the inequality $g - 3 > \lfloor g/2 \rfloor - 1$ is strict.

6 Formalization

The Formal Conjectures project [10] maintains Lean 4 formalizations of open conjectures. Conjecture 141 appears in `GraphConjecture141.lean` as `conjecture141`, marked `research open` [11], in the denominator-explicit integer form

$$\lfloor \mathrm{g}(G)/2 \rfloor - 1 + \max_v \ell(v) \leq \mathrm{t}(G).$$

Here t and ℓ are the repository definitions `largestInducedTreeSize` and `indepNeighborsCard`; Mathlib girth is natural-valued and equals zero on acyclic graphs.

We have produced a complete, `sorry`-free Lean 4 proof of exactly this statement, machine-checked with Lean toolchain v4.27.0 against current Mathlib, together with the supporting API: the star construction, the maximum-induced-tree selection with prescribed vertices, the two-neighbour maximality obstruction, the tree-path girth certificate, and the three-vertex overlap bound of Theorem 1.1. The development builds on the reusable induced-tree API we contributed alongside Conjecture 143 [9]. The proof uses no `native_decide` and no additional axioms (`#print axioms` reports `propext`, `Classical.choice`, `Quot.sound`). The source is available as pull request #4454 to the Formal Conjectures repository (which also contains our formalization of Conjecture 143) and as ancillary files with this submission.

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